EcoPulse Riyadh: Impact Snapshot & Roadmap

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Mawhiba MetaMinds

Summary

In Riyadh, cool roofs beat green roofs on energy, cost and water, but at today's household tariff even a cool roof takes about 28 years to pay back. The real lever is policy, not engineering.

EcoPulse Riyadh is my working prototype: a roof-cooling simulator. A planner picks a neighbourhood type, chooses cool or green roofs, sets how many roofs to convert, and sees the effect on street heat, electricity, CO_2, cost and payback. A fourth control sets how much of the cost the government pays, so the owner's waiting time updates live, and a panel links the result to Saudi Green Initiative targets. Every input is labelled as sourced or as my assumption, and every input can be edited.

Solo project, built with the aid of AI for research, coding and writing. The idea was first entered at NextStep Hacks 2026; this prototype, its Riyadh data, the deck and this roadmap were built for M3.

Problem in numbers

INDICATOR	VALUE	SOURCE
Peak land surface temperature	51.0 °C (2018) → 54.8 °C (2022)	Imam 2023
Urban heat-island area	412.8 km² (2018) → 455.6 km² (2022)	Imam 2023
Hottest / coolest land use	Industrial / vegetated	KSU Landsat study 1985–2015
AC share of home electricity, summer	About 70%	Alshahrani & Boait 2019
Grid emissions	About 0.569 kg CO_2/kWh (2021)	Climate Scorecard

The same 2023 study found that the Green Riyadh planting programme had not yet measurably lowered surface temperatures. Trees take years to mature; roofs can change this summer.

Impact snapshot

Scenario: 1 km² of medium-density Riyadh, 35% of land under roofs, 30% of those roofs converted (105,000 m²).

RESULT PER YEAR	COOL ROOF	GREEN ROOF
Electricity saved	1,260,000 kWh	945,000 kWh
CO_2 avoided	717 t	538 t
Street air temperature, approx.	−0.24 °C	−0.18 °C
Bill savings at 0.18 SAR/kWh	226,800 SAR	170,100 SAR
Upfront cost (one time)	6.3 million SAR	36.75 million SAR
Simple payback for the owner	**about 28 years**	**about 216 years**
Price needed for a 10-year payback	0.50 SAR/kWh	3.89 SAR/kWh
Irrigation water needed	none	yes

Neither option pays back for a household at the regulated tariff, so roll-out depends on who pays: a building-code requirement for new roofs, a utility rebate funded by avoided peak demand, or a municipal programme for the hottest districts first.

What it means for Saudi Arabia

Applied to all 455.6 km² of Riyadh's hot zones, cool roofs on 30% of buildings would save about 574 million kWh and 327,000 t of CO_2 a year, 0.12% of the Saudi Green Initiative goal of 278 million t a year by 2030 (Saudipedia). Less summer air conditioning also means less crude burned for power, which reached 0.9 million barrels a day at the July peak (US EIA, 2014). If the government pays half, owners wait 14 years instead of 28.

Scalability

EcoPulse scales by changing inputs, not code: every result comes from one small model with nine editable inputs.

- **Other districts:** set the area and roof share for any neighbourhood. Riyadh's heat-island area alone covers about 455 km².
- **Other cities:** swap in the local tariff, grid emissions and cooling demand. Jeddah, Dammam and other GCC cities fit the same hot-arid model.
- **Other interventions:** cool pavements and tree shade can be added using the same inputs-with-sources pattern.
- **Other users:** the same tool serves city planners, building owners and students, with built-in pages that explain how to use it in plain language.

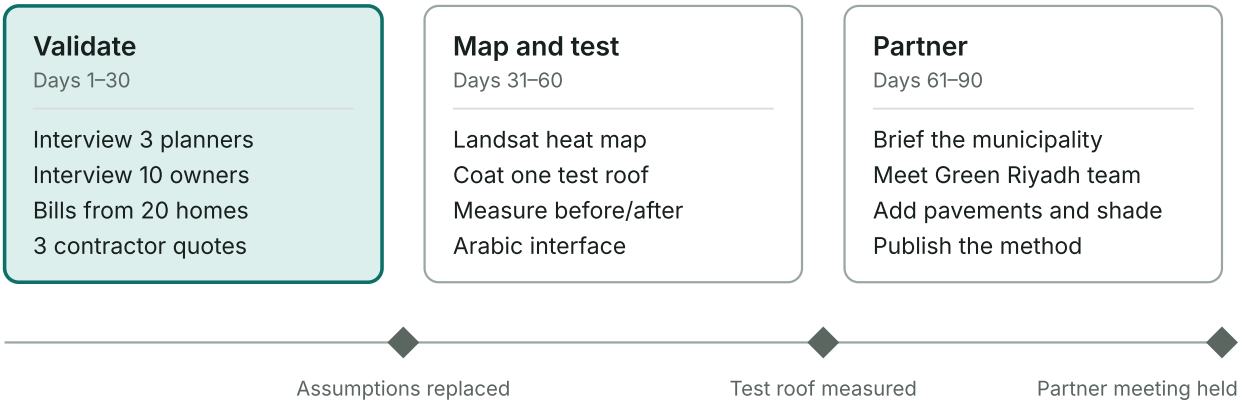
Assumptions and limits

INPUT	VALUE	TYPE
Cooling energy cut: cool / green roof	10% / 7.5%	Sourced (Elnabawi & Saber 2023)
Street air cooling at full conversion: cool / green	0.8 °C / 0.6 °C	Sourced (same study)
Household tariff	0.18 SAR/kWh	Sourced (SEC tariff)
Grid emissions	0.569 kg CO_2/kWh	Sourced (Climate Scorecard)
Cooling electricity per m² of roof	120 kWh/m²/yr	My assumption
Roof share by neighbourhood	25–50%	My assumption
Cost: cool / green roof	60 / 350 SAR/m²	My assumption

This is a screening tool, not an engineering design. Results scale linearly from a simulation study, while real cooling depends on building height, insulation and wind. The 30-day plan below replaces my three assumptions with real data first.

30-60-90 day roadmap

The first 30 days replace my three biggest assumptions with real Riyadh data; nothing scales until that gate is passed.



The test roof in days 31–60 is the first real measurement: surface and indoor temperature on one coated roof, compared with an untreated neighbour.

Sources

1. Imam, A. (2023). Examining the impact of Green Riyadh Project on alleviating urban heat island effects. J. Umm Al-Qura Univ. Eng. Archit., 14(4), 201–211. doi:10.1007/s43995-023-00030-4
2. Land use and land surface temperature in Riyadh, 1985–2015. ISPRS Archives XLII-3/W2, 2017. isprs-archives.copernicus.org
3. Elnabawi, M. H. & Saber, E. (2023). A numerical study of cool and green roof strategies in a hot arid climate. J. Building Performance Simulation, 16(1), 72–89. research.uaeu.ac.ae
4. Alshahrani, J. & Boait, P. (2019). Reducing high energy demand associated with air-conditioning needs in Saudi Arabia. Energies, 12(1), 87. mdpi.com
5. Saudi Electricity Company emissions intensity. Climate Scorecard, 2023. climatescorecard.org
6. Saudi Arabia electricity tariff schedule. ESMAP RISE, 2022. rise.esmap.org